The Surplus From a Sale and Who Captures It

ECON 201: Principles of Microeconomics
Lecture 10

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Last Class: Choosing Price and Quantity

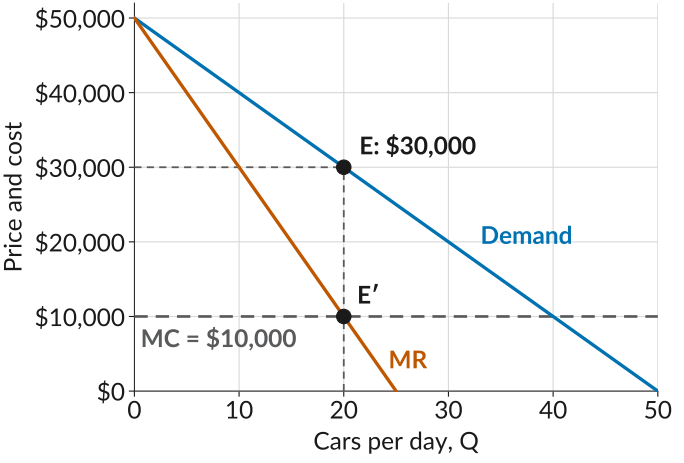
- **Marginal revenue** (MR) is the increase in revenue when one more unit is sold. To sell it, the firm lowers the price on every unit, so MR is below the price.
- **Marginal cost** (MC) is the increase in cost when one more unit is produced.
- While MR is above MC, one more unit adds to profit. Once MR falls below MC, it lowers profit. So the firm produces where MR $=$ MC.
- The demand curve then gives the price it can charge for that quantity.

Today: when something is bought and sold, how much does everyone gain from the sale?

Beautiful Cars

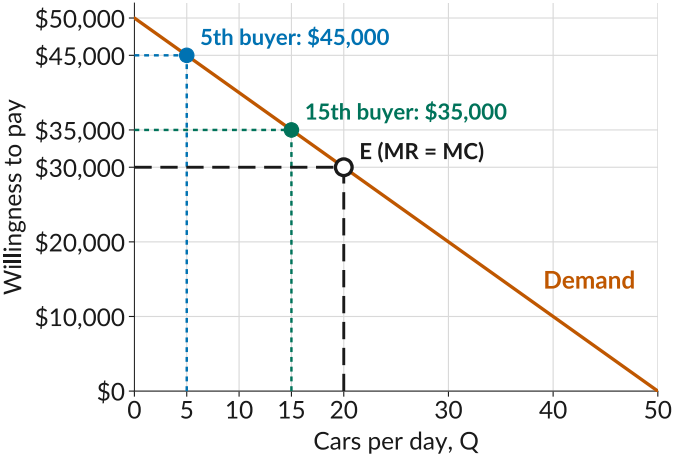
- Each car costs Beautiful Cars $10,000 to make.
- MR equals MC at 20 cars, at E'.
- At 20 cars, the demand curve gives a price of $30,000, at E.

Who gains from these sales, and by how much?



Demand as Willingness to Pay

- The height of the demand curve at Q is the willingness to pay (WTP) of the Qth buyer.
- Consider the 5th buyer. What is the total gain from the car sold to this buyer at a price of $30,000?



The Surplus on One Car

The 5th buyer's willingness to pay for a car is \$45,000. What is the **buyer's gain** from the sale?

$$\text{Buyer's gain} = \text{WTP} - P = 45{,}000 - 30{,}000 = 15{,}000$$

What is the **firm's gain** from the sale? It is the profit on this one car:

$$\text{Firm's gain} = P - \text{MC} = 30{,}000 - 10{,}000 = 20{,}000$$

The **joint surplus** (a measure of the gains from trade) is the sum of the economic rents of all involved in an economic interaction:

$$\text{Joint surplus} = \underbrace{(\text{WTP} - P)}_{\text{buyer's gain}} + \underbrace{(P - \text{MC})}_{\text{firm's gain}} = \text{WTP} - \text{MC} = 35{,}000$$

The Surplus on Two Cars

Doing the same for the 15th buyer, whose willingness to pay is $35,000:

Buyer	WTP	Joint surplus, WTP − MC	Buyer's gain, WTP − P	Firm's gain, P − MC
5th	45,000	35,000	15,000	20,000
15th	35,000	25,000	5,000	20,000

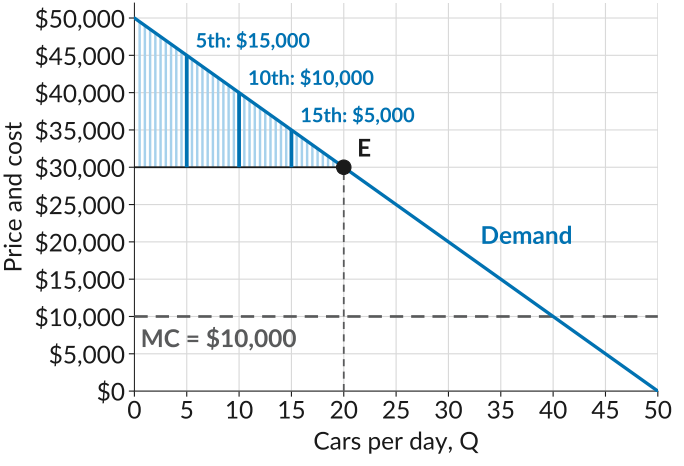
- Every buyer pays the same price, so the firm's gain is $20,000 on every car.
- The buyer's gain shrinks as willingness to pay falls.

What is the total gain in the market, adding up the gains of all the buyers and the firm's gains on all the cars it sells?

Consumer Surplus

- Each consumer's surplus from a sale is WTP $-$ P.
- **Consumer surplus (CS)**: the sum of these surpluses across all consumers.
- Calculate using the area of a triangle:

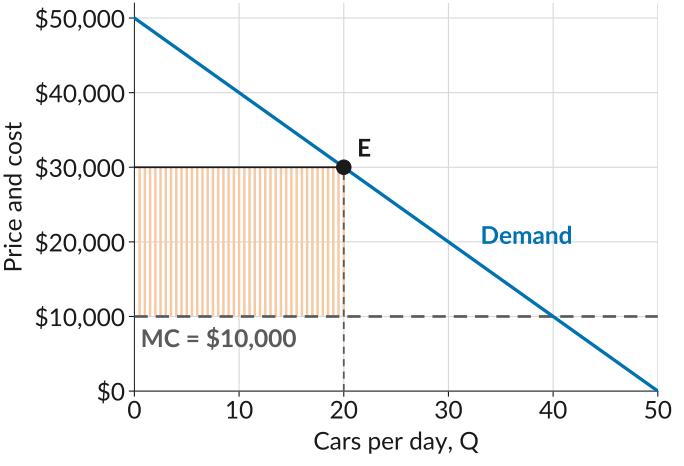
$$\begin{aligned} \text{CS} &= \tfrac{1}{2} \times \text{base} \times \text{height} \\ &= \tfrac{1}{2} \times 20 \times 20{,}000 \\ &= 200{,}000 \end{aligned}$$



Producer Surplus

- The firm's surplus from each sale is $P - \text{MC}$.
- **Producer surplus (PS)**: the sum of these surpluses across all units sold.
- Calculate using the area of a rectangle:

$$\begin{aligned} \text{PS} &= \text{base} \times \text{height} \\ &= 20 \times 20{,}000 \\ &= 400{,}000 \end{aligned}$$



Producer Surplus and Profit

Producer surplus is not the same as profit. The difference is the fixed cost.

Recall that Beautiful Cars' cost function is $C(Q) = 60{,}000 + 10{,}000Q$, with a fixed cost of \$60,000 a day. Its profit from 20 cars at \$30,000 is

$$\text{Profit} = \underbrace{30{,}000 \times 20}_{\text{Revenue}} - \underbrace{(60{,}000 + 10{,}000 \times 20)}_{\text{Cost}} = 340{,}000$$

The producer surplus was \$400,000, so

$$\text{Profit} = \text{Producer surplus} - \text{Fixed cost} = 400{,}000 - 60{,}000 = 340{,}000$$

Producer surplus compares selling cars with selling none. Even if Beautiful Cars sold no cars today, it would still pay \$60,000 for its factory, so that cost is not part of what its sales add.

Who Captures the Surplus?

The joint surplus from a sale does not depend on the price:

$$\text{Joint surplus} = \text{WTP} - \text{MC}$$

- The price P decides the split: the buyer gets WTP $- P$ and the firm gets $P -$ MC.
- The split depends on **bargaining power**: each side's ability to push the price in its favor.
- Beautiful Cars is the only seller of its car, so it has **market power**: it can set a high price, and buyers who value the car highly will still pay it.
- One buyer cannot bargain for a better deal, because the firm has many other customers. The firm takes $400,000 of the $600,000.